Table 1. Data sources and their roles

Dataset / endpoint	Cases	Controls	Role
eQTLGen blood cis-eQTL	—	—	Exposure; up to 31,684 participants (median 28,092 per variant); predominantly European
BBJ GD (GCST90018627)	2,809	172,656	Discovery; East Asian
FinnGen GD (E4_GRAVES_STRICT)	3,962	496,386	Post hoc same-disease comparison; Finnish
UKB hyperthyroidism (GCST90038636)	3,731	480,867	Broader-phenotype comparison; self-reported hyperthyroidism or thyrotoxicosis; ancestry not established as exclusively European
FinnGen GO (GRAVES_OPHT)	858	499,490	Original ophthalmopathy (TED-enriched) comparison; Finnish
FinnGen GO strict (E4_GRAVES_OPHT_STRICT)	753	499,595	Post hoc ophthalmopathy definition sensitivity; Finnish
GTEx v8 tissue eQTL	—	—	Post hoc tissue extension (thyroid, cultured fibroblasts, subcutaneous and visceral adipose, whole blood; n = 469–670; Table S4)
1000 Genomes Phase 3 European reference (n = 503)	—	—	LD clumping of instruments and reference allele frequencies

The three FinnGen R12 endpoints share participants and are not independent replications. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. UKB hyperthyroidism is broader than GD. ‘Strict’ is the FinnGen endpoint label, not proof of perfect specificity. The descriptive orbital RNA-sequencing series (four TED patients, one control) is separate from these datasets. eQTL, expression quantitative trait locus; GTEx, Genotype-Tissue Expression; LD, linkage disequilibrium; TED, thyroid eye disease; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table 2. MR and colocalization results for *TSHR*, *IGF1R* and *CTLA4* across five disease datasets

Gene	Dataset	IVs	OR (95% CI)	*P* value	Q-test *P*	PP.H2	PP.H3	PP.H4	PP.H4, p12 = 10^{-6}
TSHR	BBJ GD	1	0.12 (0.07–0.21)	1.09×10^{-14}*†	NA	<0.001	0.049	**0.951**	0.661
TSHR	FinnGen GD	1	0.08 (0.05–0.12)	3.19×10^{-32}*†	NA	<0.001	0.736	0.264	0.035
TSHR	UKB hyperthyroidism	1	0.09 (0.06–0.14)	8.77×10^{-28}*†	NA	<0.001	0.774	0.226	0.028
TSHR	FinnGen GO	1	0.10 (0.04–0.24)	2.82×10^{-7}*†	NA	<0.001	0.014	**0.986**	**0.875**
TSHR	FinnGen GO strict	1	0.15 (0.06–0.39)	9.02×10^{-5}*	NA	0.027	0.082	**0.891**	0.451
IGF1R	BBJ GD	4	1.56 (1.07–2.28)	0.0212*	0.902	0.690	0.236	0.073	0.008
IGF1R	FinnGen GD	3	1.07 (0.86–1.33)	0.539	0.882	0.731	0.259	0.010	0.001
IGF1R	UKB hyperthyroidism	4	1.35 (1.07–1.70)	0.0117*	0.250	0.400	0.196	0.404	0.063
IGF1R	FinnGen GO	3	1.41 (0.85–2.33)	0.182	0.309	0.623	0.346	0.032	0.003
IGF1R	FinnGen GO strict	3	1.52 (0.92–2.49)	0.0996	0.409	0.571	0.391	0.038	0.004
CTLA4	BBJ GD	1	0.18 (0.11–0.27)	5.45×10^{-15}*†	NA	<0.001	0.799	0.201	0.025
CTLA4	FinnGen GD	2	0.17 (0.08–0.36)	2.15×10^{-6}*†	0.0275	<0.001	0.746	0.254	0.033
CTLA4	UKB hyperthyroidism	2	0.21 (0.08–0.57)	0.00220*	0.00131	<0.001	0.047	**0.953**	0.672
CTLA4	FinnGen GO	2	0.17 (0.04–0.68)	0.0118*	0.0486	<0.001	0.022	**0.978**	**0.816**
CTLA4	FinnGen GO strict	2	0.17 (0.05–0.64)	0.00888*	0.0755	<0.001	0.021	**0.979**	**0.820**

MR estimates are ORs per approximately one SD higher genetically predicted blood expression: Wald ratio for one IV, multiplicative random-effects IVW for several. *TSHR* used one instrument, rs179252, in every dataset; its G allele raises blood expression by 0.105 SD per allele (F = 176), and its per-allele disease ORs were 0.77–0.82 (Table S3). The per-SD ORs therefore rescale a moderate per-allele association, and each single-IV *P* value equals the variant's disease-association z-test *P* value. *CTLA4* multi-IV estimates are concentrated in rs13030124, and three of the four have Q-test $P < 0.05$; in BBJ and UKB, two of four single-instrument omissions removed nominal *IGF1R* significance (Figure S2). * Nominal $P < 0.05$; † $P < 0.05/2{,}544$, the screen threshold, applied descriptively outside BBJ. PP.H2–H4 are colocalization posterior probabilities (H2, expression signal only; H3, distinct variants; H4, one shared variant) with $p1 = p2 = 10^{-4}$ and $p12 = 10^{-5}$ unless stated; bold, ≥0.80. They are not *P* values. BBJ colocalization pairs European eQTL with East Asian disease statistics. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. CI, confidence interval; eQTL, expression quantitative trait locus; IV, instrumental variable; IVW, inverse-variance weighted; MR, Mendelian randomization; NA, not applicable (one IV); OR, odds ratio; PP, posterior probability; SD, standard deviation; TED, thyroid eye disease; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table 3. Thirteen BBJ discoveries and their FinnGen GD comparison

BBJ discovery gene	Chr	IVs, BBJ/FinnGen GD	BBJ OR (95% CI)	BBJ *P*	FinnGen GD OR (95% CI)	FinnGen GD *P*	Focused support	FinnGen GD PP.H4
HLA-A	6 (MHC)	4/4	1.98 (1.73–2.27)	2.60×10^{-23}	1.08 (0.77–1.53)	0.648	No	<0.001
HLA-DQA2	6 (MHC)	1/1	2.39 (1.95–2.93)	8.30×10^{-17}	1.90 (1.64–2.20)	1.41×10^{-17}	Yes	<0.001
CTLA4	2	1/2	0.18 (0.11–0.27)	5.45×10^{-15}	0.17 (0.08–0.36)	2.15×10^{-6}	Yes	0.254
TSHR	14	1/1	0.12 (0.07–0.21)	1.09×10^{-14}	0.08 (0.05–0.12)	3.19×10^{-32}	Yes	0.264
C4A	6 (MHC)	1/3	0.45 (0.35–0.57)	2.23×10^{-10}	0.41 (0.26–0.63)	4.97×10^{-5}	Yes	<0.001
HSD3B7	16 (16p11.2 cluster)	1/2	0.29 (0.18–0.47)	2.04×10^{-7}	0.90 (0.72–1.13)	0.372	No	0.011
TUBB	6 (MHC)	2/3	1.49 (1.28–1.74)	3.14×10^{-7}	3.28 (2.09–5.17)	2.68×10^{-7}	Yes	<0.001
VKORC1	16 (16p11.2 cluster)	1/0	0.13 (0.06–0.29)	6.35×10^{-7}	NA	NA	NA	0.011
TNFSF14	19	1/1	0.63 (0.52–0.76)	1.48×10^{-6}	0.77 (0.65–0.92)	0.00361	Yes	0.303
PRSS36	16 (16p11.2 cluster)	1/1	6.00 (2.87–12.57)	2.02×10^{-6}	1.40 (1.00–1.96)	0.0490	No	0.063
MAPKAPK5	12	4/4	0.977 (0.967–0.987)	5.28×10^{-6}	0.990 (0.929–1.055)	0.751	No	0.041
PSMB8	6 (MHC)	1/1	1.25 (1.13–1.38)	6.86×10^{-6}	0.96 (0.85–1.08)	0.512	No	<0.001
IFNGR1	6	1/1	2.10 (1.51–2.91)	9.41×10^{-6}	1.62 (1.18–2.24)	0.00330	Yes	0.313

Genes passed the BBJ screen threshold ($P < 0.05/2{,}544$) and are listed by BBJ *P* value. Focused support in FinnGen GD requires the same direction and $P < 0.05/13$; 7 of 13 qualify. IVs are counted separately for each dataset because instrument availability differs. The five MHC genes and the three chromosome 16p11.2 genes are not independent signals; the latter are compatible with one regional association. *MAPKAPK5* is shown to three decimals to preserve its small effect. NA, not estimable (no retained instrument), not a negative result; *VKORC1* colocalization does not require an MR instrument. PP.H4, shared-variant posterior probability under $p12 = 10^{-5}$; bold, ≥0.80. Chr, chromosome of the strongest instrument; MHC, major histocompatibility complex; CI, confidence interval; IV, instrumental variable; MR, Mendelian randomization; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Figure 1

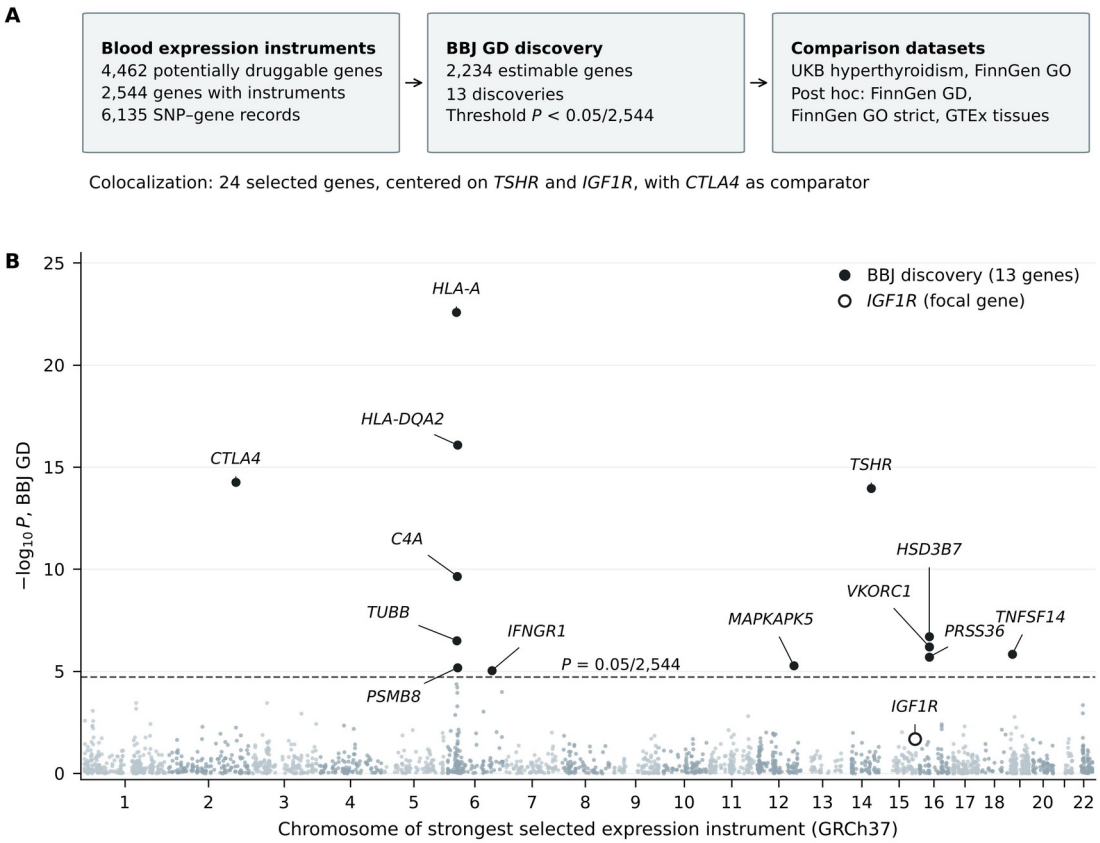


Figure 1. Study design and gene-level discovery in BBJ. (A) Analysis flow. Blood expression instruments from eQTLGen were tested against BBJ GD for discovery; UKB hyperthyroidism and FinnGen GO were the original comparison datasets, and FinnGen GD, FinnGen GO strict and the GTEx tissue extension were added post hoc. The three FinnGen endpoints overlap and are not independent replications. (B) Each grey point is one estimable gene (n = 2,234), placed at the GRCh37 position of its strongest selected expression instrument and plotted by its BBJ MR *P* value; points are genes, not single variants. Filled black points are the 13 genes that passed $P < 0.05/2{,}544$ (dashed line, $-\log_{10}P \approx 4.71$); the open circle is the focal gene *IGF1R*, which did not. The threshold applies to the BBJ discovery screen, not to later exploratory analyses. GTEx, Genotype-Tissue Expression; MR, Mendelian randomization; SNP, single-nucleotide polymorphism; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Figure 2

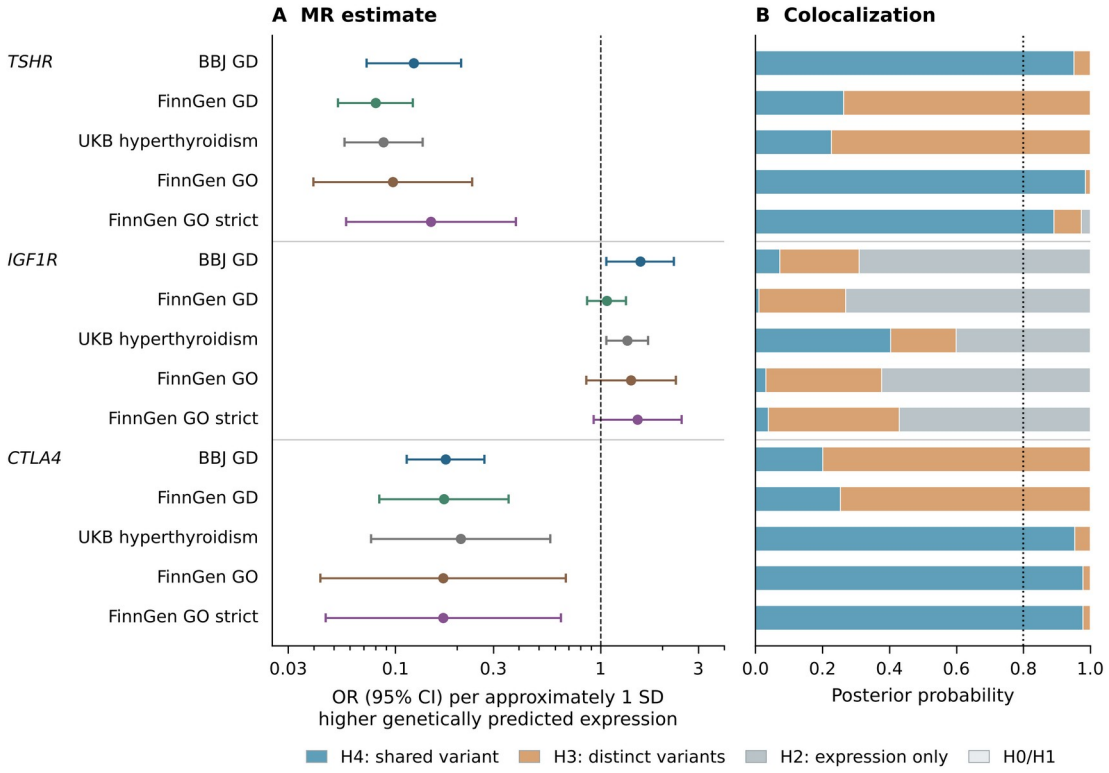


Figure 2. *TSHR*, *IGF1R* and *CTLA4* across five disease datasets. (A) MR odds ratios with 95% CIs per approximately 1 SD higher genetically predicted blood expression: Wald ratio for one instrument, multiplicative random-effects IVW for several. The axis is logarithmic and the dashed line is OR = 1. *TSHR* used one instrument, rs179252, in every dataset. Colours identify datasets. (B) Colocalization posterior probabilities for the same comparisons ($p1 = p2 = 10^{-4}$, $p12 = 10^{-5}$), stacked from zero as H4 (one shared variant), H3 (distinct variants), H2 (expression signal only) and H0/H1 (neither trait, or disease only). The dotted line marks PP.H4 = 0.80, the descriptive reporting threshold. Estimates, *P* values and posterior probabilities are given in Table 2; *P* values and posterior probabilities answer different questions. FinnGen GO controls include people with GD but no recorded eye disease. CI, confidence interval; IVW, inverse-variance weighted; MR, Mendelian randomization; OR, odds ratio; PP, posterior probability; SD, standard deviation; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Figure 3

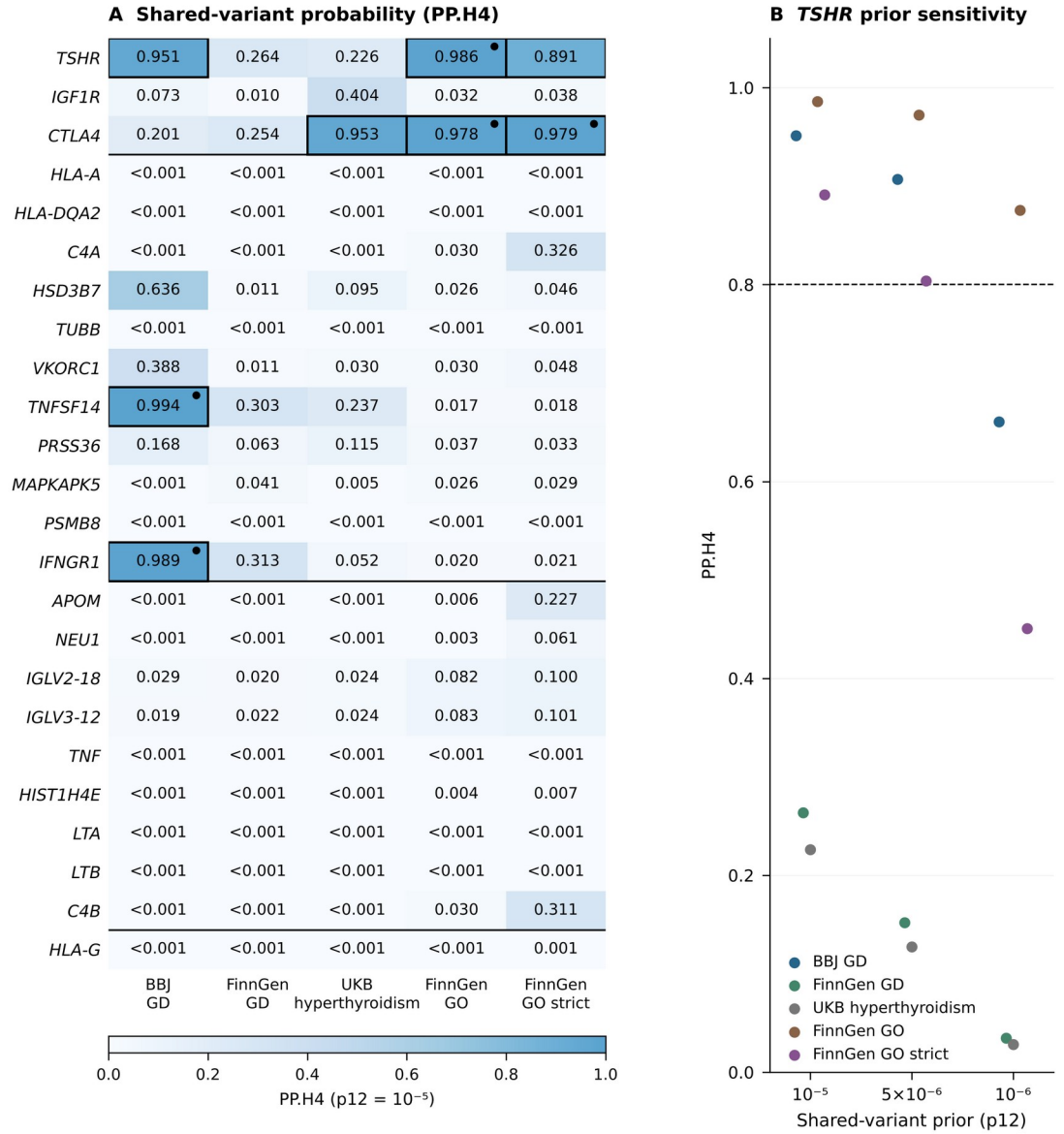


Figure 3. Colocalization across 24 selected genes and *TSHR* prior sensitivity. (A) PP.H4 for each gene and dataset (p1 = p2 = 10^{-4}, p12 = 10^{-5}); shading follows the colour bar. Boxed cells have PP.H4 ≥ 0.80, and a dot marks cells that remain ≥ 0.80 at p12 = 10^{-6}. Horizontal lines separate *TSHR*, *IGF1R* and *CTLA4*; the other 11 BBJ discoveries; the nine additional FinnGen GD discoveries; and *HLA-G*, added because its MR result changed with the allele-frequency source. Only these 24 genes underwent colocalization. Values below 0.001 are shown as <0.001; full-precision values are in the Supplementary Dataset. Low BBJ PP.H4 for the two *IGLV* genes partly reflects absence of the lead expression variant from the BBJ data. (B) *TSHR* PP.H4 in each dataset at three shared-variant priors (p12); points are offset slightly to avoid overlap, and the dashed line marks 0.80. PP.H4 is a posterior probability under the colocalization model, not a *P* value. MR, Mendelian randomization; PP.H4, posterior probability of one shared causal variant; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table S1. Additional FinnGen GD screen discoveries

Gene	Chr	FinnGen GD IVs	FinnGen GD OR (95% CI)	FinnGen GD *P*	BBJ PP.H4	FinnGen GD PP.H4	FinnGen GO strict PP.H4
APOM	6 (MHC)	1	0.05 (0.04–0.06)	1.43×10^{-108}	<0.001	<0.001	0.227
NEU1	6 (MHC)	1	0.04 (0.03–0.06)	1.43×10^{-108}	<0.001	<0.001	0.061
IGLV2-18	22	1	0.56 (0.48–0.66)	7.52×10^{-12}	0.029	0.020	0.100
IGLV3-12	22	1	1.20 (1.14–1.26)	7.52×10^{-12}	0.019	0.022	0.101
TNF	6 (MHC)	1	0.60 (0.51–0.71)	4.65×10^{-10}	<0.001	<0.001	<0.001
HIST1H4E	6 (MHC)	1	2.54 (1.88–3.43)	1.51×10^{-9}	<0.001	<0.001	0.007
LTA	6 (MHC)	1	0.46 (0.35–0.61)	3.75×10^{-8}	<0.001	<0.001	<0.001
LTB	6 (MHC)	1	0.18 (0.10–0.36)	4.89×10^{-7}	<0.001	<0.001	<0.001
C4B	6 (MHC)	2	1.63 (1.31–2.03)	1.10×10^{-5}	<0.001	<0.001	0.311

Genes outside the BBJ discovery set that passed $P < 0.05/2{,}544$ in the post hoc full FinnGen GD screen (1000 Genomes European reference frequencies). These are not independent of the screen and seven of nine are in the MHC. *APOM*/*NEU1* share rs3101018 and *IGLV2-18*/*IGLV3-12* share rs5759410. The lead *IGLV* expression variant is absent in BBJ, weakening interpretation of its low PP.H4. None reached PP.H4 ≥0.80 ($p12 = 10^{-5}$) in any dataset. Chr, chromosome; CI, confidence interval; IV, instrumental variable; MHC, major histocompatibility complex; OR, odds ratio; PP.H4, shared-variant posterior probability; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table S2. Frequency-sensitive exploratory candidates

Gene	Dataset	Frequency source	IVs	OR (95% CI)	*P* value	Q-test *P*
C4B	FinnGen GD	1000G EUR	2	1.63 (1.31–2.03)	1.10×10^{-5}	7.79×10^{-7}
C4B	FinnGen GD	eQTLGen cohort	2	1.75 (1.31–2.33)	1.48×10^{-4}	1.36×10^{-8}
C4B	FinnGen GO strict	1000G EUR	2	1.68 (1.44–1.95)	2.74×10^{-11}	0.122
C4B	FinnGen GO strict	eQTLGen cohort	2	1.80 (1.46–2.23)	6.17×10^{-8}	0.0587
HLA-G	FinnGen GD	1000G EUR	16	0.86 (0.72–1.03)	0.107	9.98×10^{-24}
HLA-G	FinnGen GD	eQTLGen cohort	16	0.60 (0.47–0.75)	1.72×10^{-5}	2.26×10^{-10}
HLA-G	FinnGen GO strict	1000G EUR	16	0.91 (0.76–1.08)	0.267	0.0231
HLA-G	FinnGen GO strict	eQTLGen cohort	16	0.70 (0.53–0.92)	0.0122	0.131

Exposure effects were reconstructed with 1000 Genomes Phase 3 European reference allele frequencies (1000G EUR; primary) or eQTLGen cohort frequencies; variant sets and outcome effects were held fixed, so changes reflect rescaled expression effects and IVW weights. *HLA-G* was added because its eQTLGen-frequency *P* crossed the screen threshold ($P < 0.05/2{,}544$). Q tests heterogeneity, not directional pleiotropy; its *P* value is not a gene-association *P* value. All 72 omission estimates and instrument weights are in the workbook. CI, confidence interval; IV, instrumental variable; IVW, inverse-variance weighted; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table S3. *TSHR* regional colocalization and prior sensitivity

Dataset	Matched variants	Disease lead (*P*)	rs179252 *P*	rs179252-G OR (95% CI)	H4 top variant	PP.H4, 10^{-5} / 10^{-6}	PP.H4 range across windows
BBJ GD	4,336	rs28414437 (1.83×10^{-15})	1.09×10^{-14}	0.80 (0.76–0.85)	rs179252	**0.951** / 0.661	0.951–0.951
FinnGen GD	6,579	rs1023586 (1.58×10^{-36})	3.19×10^{-32}	0.77 (0.73–0.80)	rs1023586	0.264 / 0.035	0.264–0.264
UKB hyperthyroidism	5,667	rs4903961 (1.17×10^{-43})	8.77×10^{-28}	0.77 (0.74–0.81)	rs1023586	0.226 / 0.028	0.226–0.226
FinnGen GO	6,579	rs3783947 (1.23×10^{-7})	2.82×10^{-7}	0.78 (0.71–0.86)	rs179252	**0.986** / **0.875**	0.986–0.987
FinnGen GO strict	6,579	rs2110695 (1.34×10^{-5})	9.02×10^{-5}	0.82 (0.74–0.90)	rs179252	**0.891** / 0.451	0.891–0.905

Disease lead, the strongest disease-associated variant among the matched cis variants, with its *P* value; rs179252 *P*, z-test *P* of the aligned G-allele disease association (the UKB source BOLT-LMM *P* was 1.3×10^{-27}). H4 top variant, the variant with the highest posterior under the shared-variant hypothesis; it is not necessarily the disease lead. PP.H4 under p12 = 10^{-5} and 10^{-6} (p1 = p2 = 10^{-4}); bold, ≥0.80. Window range covers the full source cis-region, ±500 kb, ±250 kb and the five-outcome intersection at p12 = 10^{-5}; it is not a confidence interval. All 240 window settings are in TSHR_windows. BBJ pairs European eQTL with East Asian disease statistics. CI, confidence interval; eQTL, expression quantitative trait locus; OR, odds ratio; PP.H4, shared-variant posterior probability; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table S4. *TSHR* and *IGF1R* tissue expression extension

A. Availability

Gene / tissue	GTEx N	Instrument (F)	Status
TSHR / Thyroid	574	—	No instrument (minimum eQTL $P = 9.60\times10^{-8}$)
IGF1R / Thyroid	574	rs1815009 (F = 61.1)	Estimable
TSHR / Cultured fibroblasts	483	—	Not tested by GTEx in this tissue
IGF1R / Cultured fibroblasts	483	rs6598538 (F = 33.8)	Estimable
TSHR / Subcutaneous adipose	581	rs12101255 (F = 38.5)	Estimable
IGF1R / Subcutaneous adipose	581	—	No instrument (minimum eQTL $P = 3.56\times10^{-7}$)
TSHR / Visceral adipose	469	rs74064796 (F = 32.1)	Estimable
IGF1R / Visceral adipose	469	—	No instrument (minimum eQTL $P = 7.76\times10^{-6}$)
TSHR / Whole blood	670	—	No instrument (minimum eQTL $P = 2.46\times10^{-4}$)
IGF1R / Whole blood	670	—	No instrument (minimum eQTL $P = 5.42\times10^{-5}$)

B. Estimates

Gene / tissue	Dataset	Family	OR (95% CI)	*P* value	Below family threshold	Direction vs blood
TSHR / Subcutaneous adipose	BBJ GD	Secondary	0.46 (0.38–0.55)	1.92×10^{-15}	Yes	Same
TSHR / Subcutaneous adipose	FinnGen GD	Primary	0.36 (0.31–0.43)	1.28×10^{-34}	Yes	Same
TSHR / Subcutaneous adipose	UKB hyperthyroidism	Secondary	0.31 (0.27–0.37)	2.86×10^{-42}	Yes	Same
TSHR / Subcutaneous adipose	FinnGen GO	Secondary	0.45 (0.32–0.63)	5.76×10^{-6}	Yes	Same
TSHR / Subcutaneous adipose	FinnGen GO strict	Secondary	0.51 (0.35–0.74)	3.67×10^{-4}	Yes	Same
TSHR / Visceral adipose	BBJ GD	Secondary	0.56 (0.46–0.67)	2.64×10^{-10}	Yes	Same
TSHR / Visceral adipose	FinnGen GD	Primary	0.50 (0.43–0.58)	4.09×10^{-19}	Yes	Same
TSHR / Visceral adipose	UKB hyperthyroidism	Secondary	0.39 (0.34–0.45)	9.19×10^{-38}	Yes	Same
TSHR / Visceral adipose	FinnGen GO	Secondary	0.69 (0.49–0.98)	0.0368	No	Same
TSHR / Visceral adipose	FinnGen GO strict	Secondary	0.78 (0.54–1.13)	0.186	No	Same
IGF1R / Thyroid	BBJ GD	Secondary	1.23 (0.90–1.68)	0.200	No	Same
IGF1R / Thyroid	FinnGen GD	Primary	0.76 (0.56–1.02)	0.0692	No	Opposite
IGF1R / Thyroid	UKB hyperthyroidism	Secondary	1.33 (0.99–1.80)	0.0594	No	Same
IGF1R / Thyroid	FinnGen GO	Secondary	1.02 (0.54–1.94)	0.945	No	Same
IGF1R / Thyroid	FinnGen GO strict	Secondary	1.23 (0.62–2.44)	0.549	No	Same
IGF1R / Cultured fibroblasts	BBJ GD	Secondary	0.91 (0.70–1.20)	0.511	No	Opposite
IGF1R / Cultured fibroblasts	FinnGen GD	Primary	1.11 (0.89–1.38)	0.359	No	Same
IGF1R / Cultured fibroblasts	UKB hyperthyroidism	Secondary	1.03 (0.82–1.29)	0.815	No	Same
IGF1R / Cultured fibroblasts	FinnGen GO	Secondary	0.78 (0.49–1.25)	0.301	No	Opposite
IGF1R / Cultured fibroblasts	FinnGen GO strict	Secondary	0.78 (0.47–1.30)	0.341	No	Opposite

Post hoc fixed-panel extension using GTEx v8; tissue colocalization was not performed. A, availability of the ten gene–tissue combinations; F statistics are for the single retained instrument. ‘Not tested’ means the gene is not among the genes GTEx tested in that tissue. GTEx whole blood (n = 670) yielded no instrument at the fixed threshold, whereas the main analysis used eQTLGen (up to 31,684 participants). B, all 20 estimates. The primary family is the ten FinnGen GD comparisons ($P < 0.005$); the secondary family is the 40 comparisons with the other datasets ($P < 0.00125$); unestimable combinations stay in both denominators. ORs are per unit of inverse-normal-transformed tissue expression and are not comparable in size with the blood ORs. With one instrument, each *P* value equals that variant’s disease-association z-test *P* value (UKB source BOLT-LMM *P* values differ slightly). The adipose *TSHR* instruments (rs12101255, rs74064796; 50 bp apart in *TSHR* intron 1) are themselves blood *TSHR* eQTLs near rs179252 and re-express the regional association rather than providing independent evidence. Cultured fibroblasts and adipose tissue are not orbital tissue. Direction, compared with the blood estimate in the same dataset. CI, confidence interval; eQTL, expression quantitative trait locus; GTEx, Genotype-Tissue Expression; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Figure S1

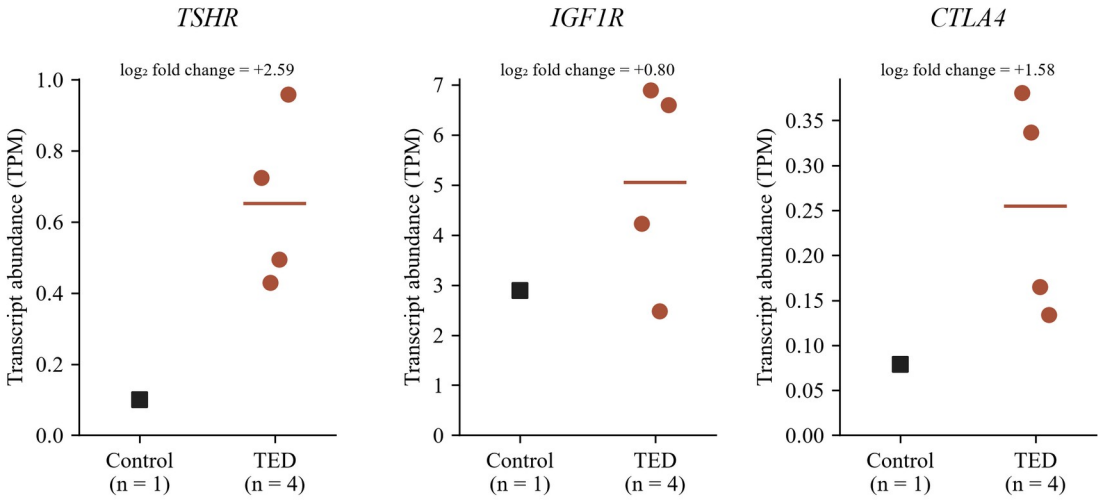


Figure S1. Descriptive orbital expression of *TSHR*, *IGF1R* and *CTLA4*. Transcript abundance in orbital adipose tissue collected during orbital decompression from four patients with inactive TED, and in nasal fat pad from one person without thyroid disease undergoing cosmetic upper blepharoplasty. Points are biological samples (technical replicates combined); horizontal lines show TED means. The $\log_2$ fold change is $\log_2$[(mean TED TPM + 0.01)/(control TPM + 0.01)]. Vertical axes differ by gene. The control came from a different fat depot, and one control cannot estimate control-group variance; no inferential test was performed, and these observations do not validate the genetic findings. TED, thyroid eye disease; TPM, transcripts per million.

Figure S2

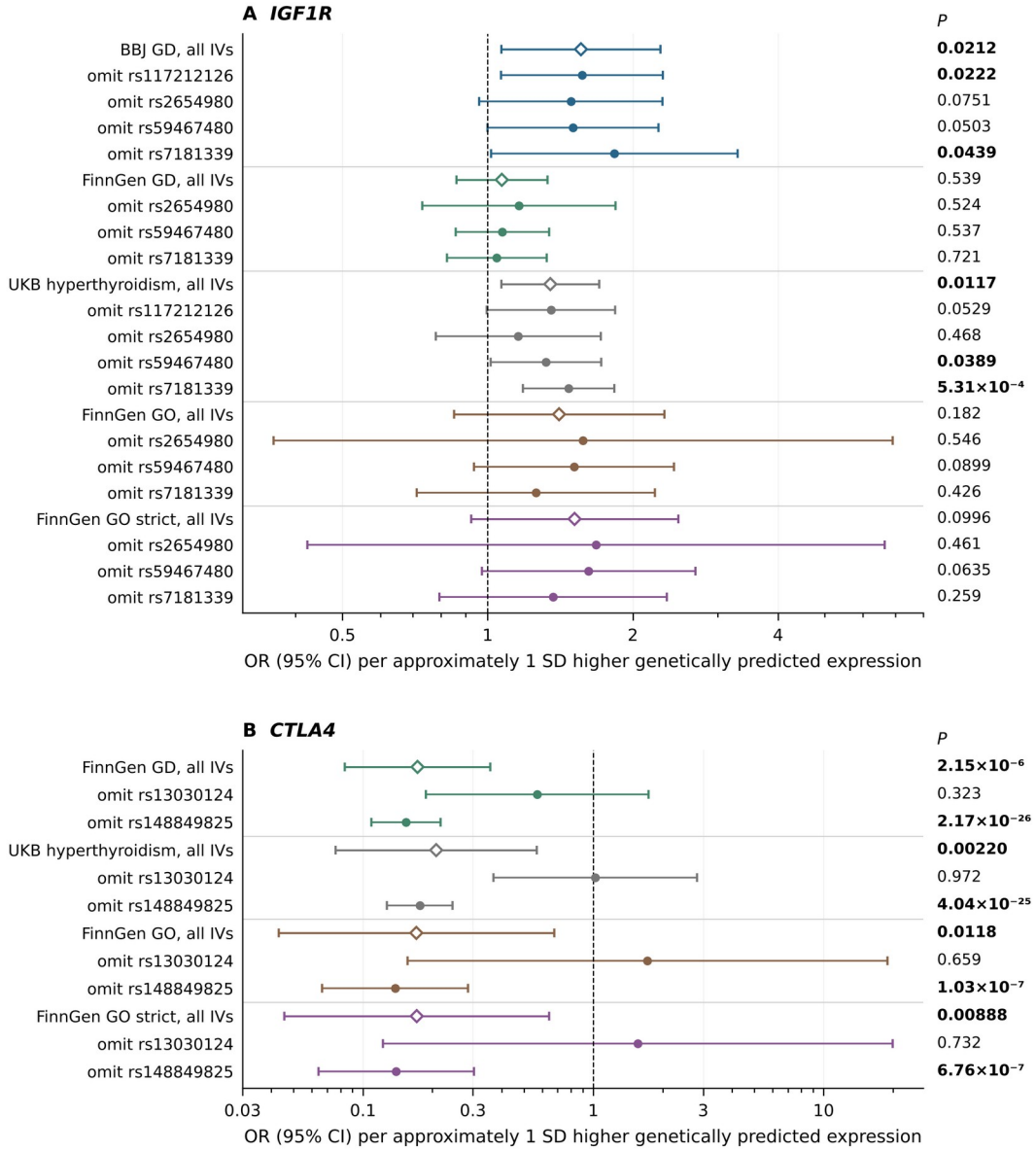


Figure S2. Dependence of *IGF1R* and *CTLA4* estimates on individual instruments. For each dataset, the open diamond is the estimate using all instruments (as in Table 2), and the filled points below it omit the named instrument (IVW when at least two instruments remain, Wald ratio when one remains). Lines are 95% CIs on a logarithmic axis; the dashed line is OR = 1. *P* values are shown at right; bold, nominal $P < 0.05$. (A) *IGF1R* in all five datasets. (B) *CTLA4* in the four FinnGen and UKB datasets; BBJ *CTLA4* and all *TSHR* estimates use one instrument and could not be assessed. The 25 post hoc omission estimates reuse the same association data and are not independent replications; loss of nominal significance can reflect reduced precision as well as the omitted variant's contribution. CI, confidence interval; IV, instrumental variable; IVW, inverse-variance weighted; OR, odds ratio; SD, standard deviation; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.